

Santosh Bala N

9884515122 | santmartian@gmail.com | [LinkedIn](#) | [GitHub](#)

EDUCATION

Vellore Institute of Technology

B.Tech Computer Science and Engineering - CGPA: 8.7

Vellore, TN

2024 – 2028

Vidya Mandir Senior Secondary School

High School Diploma - 93.4%

Chennai, TN

2010 – 2024

EXPERIENCE

Google Summer of Code 2026

Remote

Open Source Contributor - PSLab Project under FOSSASIA

May 2026 – Present

- Writing firmware in **C** with a 3-layer system architecture (platform / system / application) for a new open-source lab instrument, enabling high-throughput data capture with clean separation of hardware abstraction and application logic.
- Implementing a **logic analyzer** at **200 MSPS** (million samples/second) and oscilloscope at **80 MSPS** with industry standard instrument control protocol **SCPI** command interface over USB and **wireless UDP based streaming** for untethered operation.
- Validating end-to-end system integration with PSLab mobile and desktop applications.

Team Vyadh - Mars Rover Team, VIT

Vellore, TN

Software Lead

April 2025 – Present

- Lead a multidisciplinary team of engineers, owning the rover's software and firmware architecture end-to-end; redesigned a fragmented control system into a unified, deterministic architecture with closed-loop velocity control and fault tolerance.
- Architected the fully autonomous outdoor navigation stack in **Python** using **ROS 2**; wrote a custom **C++ costmap plugin** for dynamic polygon-based keep-out zones.
- Designed a real-time telemetry pipeline over **MQTT** streaming GPS/odometry, battery, and sensor data; built a 6 channel HD camera streaming pipeline in **GStreamer** with hardware accelerated encoding and watchdog monitored auto-reconnect, achieving **100%** streaming uptime across competition.

Antropi Robotics

Bengaluru, KA

Software Intern

May 2026 – June 2026

- Built a **6-DoF pose estimation** pipeline (**FoundationPose**, **SAM3**) in Python for a KUKA robotic arm enabling arbitrary object pick-and-place, and tuned an AMR navigation/docking stack (ROS 2) achieving **sub-1 cm** docking accuracy.
- Reverse-engineered proprietary CAD analysis features and built equivalent **automated manufacturing quoting** tooling in Python using OpenCascade.

PROJECTS

Hermes [GitHub] | Go, Redis Streams, Prometheus, Grafana, Docker

2026

- Built a pub-sub middleware in **Go** inspired by ROS 2/DDS internals; implements broker-managed topics, reliable **TCP** and best-effort **UDP** transports, **Redis Streams** durability with offset-based replay, JSON schema enforcement, node discovery, and **Prometheus**/Grafana observability.
- Used **OpenAI Codex** to scaffold the initial test suite, then hand wrote edge case tests for backpressure and dead letter queue handling.
- Sustained **256,000/256,000** reliable deliveries across **512 subscribers** with **48 ms p99 latency** and zero drops;

Multi Node Telemetry Pipeline | Go, Kafka, Kubernetes, Terraform, Prometheus, Grafana

2026

- Built a distributed telemetry ingestion pipeline simulating multiple nodes, with **Kafka** based ingestion across partitioned and horizontally scalable consumers; real-time anomaly detection on sensor streams.
- Deployed on **Kubernetes** via **Terraform**-provisioned infrastructure; built **Prometheus**/Grafana dashboards for ingestion and system health monitoring.

Distributed Job Scheduler | Go, Raft, gRPC, etcd, Kubernetes

2026

- Built a fault-tolerant distributed task scheduler implementing **Raft consensus** for leader election and coordinator failover, with idempotent execution and dead-letter handling for failing jobs.
- Exposed a **gRPC** API for job submission and status querying; tested leader/worker failover and autoscaled workers on **Kubernetes** based on queue depth.

TECHNICAL SKILLS

Languages: C, C++, Python, Go, JavaScript

Systems & Distributed: Redis Streams, Kafka, MQTT, Prometheus, Grafana, REST, Docker

Cloud & Infra: Kubernetes, Terraform, gRPC, Raft/etcd, GitHub Actions

Robotics & Perception: ROS 2, Nav2, GStreamer, FoundationPose, SAM3, ArUco

Embedded: STM32, ESP32, RP2350, PIO, DMA, CAN

Tools: Git, Linux, CMake, GDB

AI Tools: Claude, OpenAI Codex, Github Copilot